



Exploratory Data Analysis (EDA) Report



Project Objective

The objective of this project is to perform **Exploratory Data Analysis (EDA)** on the **Zomato Restaurant Dataset** to uncover meaningful insights about restaurant performance, customer preferences, pricing trends, online ordering services, and location-based patterns.

The analysis helps transform raw restaurant data into actionable business intelligence that can support better decision-making for restaurant owners, managers, and stakeholders.



Dataset Overview

The Zomato dataset contains detailed information about restaurants, including:

- Restaurant Name
- Restaurant Type
- Location
- Online Order Availability
- Table Booking Availability
- Customer Ratings
- Customer Votes
- Approximate Cost for Two
- Popular Dishes
- Cuisines
- Reviews
- Menu Information

The dataset was analyzed using Python libraries to identify hidden trends and valuable business insights.



Data Preprocessing

Before analysis, the dataset was cleaned to improve accuracy and reliability.

Data Cleaning Performed

- ✓ Missing values identified and handled
 - ✓ Duplicate records removed
 - ✓ Data types corrected
 - ✓ Outliers detected using Boxplots
 - ✓ Outliers treated using the IQR Method
 - ✓ Dataset prepared for visualization and analysis
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Exploratory Data Analysis

The following analyses were performed:

- Dataset Overview
 - Missing Value Analysis
 - Outlier Detection
 - Univariate Analysis
 - Bivariate Analysis
 - Correlation Analysis
 - Feature Engineering
 - Business Insights
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Key Business Insights

Customer Behaviour

- Restaurants with a higher number of customer votes generally receive higher ratings.
- Restaurants offering online ordering are more common, indicating strong customer demand for convenience.
- Restaurant popularity varies significantly across different locations.

Pricing Analysis

- Most restaurants belong to the medium price range.
- Higher pricing does not necessarily result in higher customer ratings.

- Pricing should be balanced with service quality to maximize customer satisfaction.

Restaurant Performance

- Highly rated restaurants attract greater customer engagement.
- Restaurant type influences customer popularity and demand.
- Customer satisfaction is affected by multiple factors, including food quality, pricing, and service.

Data Quality

- Outlier treatment improved the quality and reliability of statistical analysis.
- Clean data resulted in more accurate visualizations and business conclusions.



Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Improve food quality to achieve higher customer ratings.
- Expand online ordering services to improve customer convenience.
- Encourage customers to leave ratings and reviews.
- Maintain competitive pricing based on market demand.
- Focus marketing campaigns on high-demand locations.
- Invest in restaurant categories with consistently strong customer engagement.
- Regularly monitor customer feedback to improve service quality.
- Use data-driven decision making for future business expansion.



Technologies Used

Technology	Purpose
Python	Data Analysis
Pandas	Data Manipulation
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization

Technology	Purpose
Jupyter Notebook	Analysis Environment
VS Code	Development
Git & GitHub	Version Control

Conclusion

The **Zomato Business Insights** project demonstrates how Exploratory Data Analysis (EDA) can transform raw restaurant data into meaningful business knowledge.

The analysis revealed valuable insights regarding customer preferences, restaurant performance, pricing strategies, online ordering trends, and location-based opportunities. These findings can help restaurant owners improve customer satisfaction, optimize pricing strategies, and make informed business decisions using data-driven approaches.

Project Information

Project Title: Zomato Business Insights

Project Type: Exploratory Data Analysis (EDA)

Dataset: Zomato Restaurant Dataset

Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Git & GitHub